Chapter 21

Debug Assistant

21.1 Overview

Debug Assistant is an auxiliary module that features a set of functions to help locate bugs and issues during software debugging.

21.2 Features

- **Read/write monitoring**: Monitors whether the High-Performance dual-core CPU (HP CPU0 and HP CPU1) bus reads from or writes to a specified memory address space. A detected read or write in the monitored address space will trigger an interrupt.
- **Stack pointer (SP) monitoring**: Monitors whether the SP exceeds the specified address space. A bounds violation will trigger an interrupt.
- **Program counter (PC) logging**: Records the PC value. The developer can get the last PC value at the most recent reset of HP CPU0 or HP CPU1.
- **Bus access logging**: Records the information about bus access. When the HP CPU0, HP CPU1, or the Direct Memory Access controller (DMA) writes a specified value, the Debug Assistant module will record the data type, address of this write operation, and additionally the PC value when the write is performed by HP CPU0 or HP CPU1, and push such information to the HP L2MEM.

21.3 Functional Description

21.3.1 Region Read/Write Monitoring

The Debug Assistant module can monitor reads/writes performed by HP CPU0 bus in a certain address space, i.e., memory region. Whenever the bus reads or writes in the specified address space, an interrupt will be triggered. Please refer to Table 1.4-1 in Chapter 1 *High-Performance CPU*. In this chapter, when HP CPU0 bus accesses the data space, it is referred to as the Data bus, and when HP CPU0 bus accesses the AHB peripheral space, it is referred to as the Peripheral bus. The Data bus can monitor two memory regions (assuming they are HP CPU0 region 0 and HP CPU0 region 1, defined by developers' needs) at the same time, and so can Peripheral bus.

This is also the case for HP CPU1.

21.3.2 SP Monitoring

The Debug Assistant module can monitor the SP so as to prevent stack overflow or erroneous push/pop in HP CPU0. When the HP CPU0 SP goes below or beyond the lower or upper bound of the HP CPU0 SP monitored region, the module will record the PC pointer and generate an interrupt. The bound is configured by software.

This is also the case for HP CPU1.

21.3.3 PC Logging

In some cases, software developers want to know the PC at the last reset of HP CPU0. For instance, when the program is stuck and can only be reset, the developer may want to know where the program got stuck in order to debug. The Debug Assistant module can record the PC at the last reset of HP CPU0, which can be then read for software debugging.

This is also the case for HP CPU1.

21.3.4 CPU/DMA Bus Access Logging

The Debug Assistant module can record the information about the HP CPU0 Data bus's, HP CPU1 bus's, and DMA bus's write behaviors in real time. When a write operation occurs in or a specific value is written to a specified address space, the module will record the bus type, the address, PC (only when the write is performed by the HP CPU0 or HP CPU1, will PC be recorded), and other information, and then store the data in the HP L2MEM in a certain format. The specified address range must fall within the permitted memory regions of HP SPM or HP L2MEM.

21.4 Interrupts

The following interrupt sources from the Debug Assistant module can generate the interrupt signal ASSIST_DEBUG_INT.

- ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_INT: Triggered when the HP CPU0 Data bus reads in HP CPU0 region 0.
- ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_INT: Triggered when the HP CPU0 Data bus writes in HP CPU0 region 0.
- ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_INT: Triggered when the HP CPU0 Data bus reads in HP CPU0 region 1.
- ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_INT: Triggered when the HP CPU0 Data bus writes in HP CPU0 region 1.
- ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_INT: Triggered when the HP CPU0 Peripheral bus reads in HP CPU0 region 0.
- ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_INT: Triggered when the HP CPU0 Peripheral bus writes in HP CPU0 region 0.
- ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_INT: Triggered when the HP CPU0 Peripheral bus reads in HP CPU0 region 1.

- ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_INT: Triggered when the HP CPU0 Peripheral bus writes in HP CPU0 region 1.
- ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_INT: Triggered when the HP CPU0 SP goes below the lower bound of the HP CPU0 SP monitored region.
- ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_INT: Triggered when the HP CPU0 SP goes beyond the upper bound of the HP CPU0 SP monitored region.
- ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_INT: Triggered when the HP CPU1 Data bus reads in HP CPU1 region 0.
- ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_INT: Triggered when the HP CPU1 Data bus writes in HP CPU1 region 0.
- ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_INT: Triggered when the HP CPU1 Data bus reads in HP CPU1 region 1.
- ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_INT: Triggered when the HP CPU1 Data bus writes in HP CPU1 region 1.
- ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_INT: Triggered when the HP CPU1 Peripheral bus reads in HP CPU1 region 0.
- ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_INT: Triggered when the HP CPU1 Peripheral bus writes in HP CPU1 region 0.
- ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_INT: Triggered when the HP CPU1 Peripheral bus reads in HP CPU1 region 1.
- ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_INT: Triggered when the HP CPU1 Peripheral bus writes in HP CPU1 region 1.
- ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_INT: Triggered when the HP CPU1 SP goes below the lower bound of the HP CPU1 SP monitored region.
- ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_INT: Triggered when the HP CPU1 SP goes beyond the upper bound of the HP CPU1 SP monitored region.

Note:
For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter 12 *Interrupt Matrix* > Section 12.2 *Interrupt Terminology in ESP32-P4*.

Each interrupt source can be configured by a common set of registers that are described in Section *Interrupt Configuration Registers*. The specific registers can be found in Section 21.6 *Register Summary*.

21.5 Recommended Operation

21.5.1 Region Monitoring and SP Monitoring Configuration

The Debug Assistant module can monitor reads and writes performed by the Data bus and Peripheral bus of HP CPU0 and HP CPU1. Two memory regions on each bus can be monitored at the same time. All the monitoring modes supported by the Debug Assistant module are listed below:

- Monitoring the read/write operations performed by Data bus
  - HP CPU0 Data bus reads in HP CPU0 region 0
  - HP CPU0 Data bus writes in HP CPU0 region 0
  - HP CPU0 Data bus reads in HP CPU0 region 1
  - HP CPU0 Data bus writes in HP CPU0 region 1
  - HP CPU1 Data bus reads in HP CPU1 region 0
  - HP CPU1 Data bus writes in HP CPU1 region 0
  - HP CPU1 Data bus reads in HP CPU1 region 1
  - HP CPU1 Data bus writes in HP CPU1 region 1
- Monitoring the read/write operations performed by Peripheral bus
  - HP CPU0 Peripheral bus reads in HP CPU0 region 0
  - HP CPU0 Peripheral bus writes in HP CPU0 region 0
  - HP CPU0 Peripheral bus reads in HP CPU0 region 1
  - HP CPU0 Peripheral bus writes in HP CPU0 region 1
  - HP CPU1 Peripheral bus reads in HP CPU1 region 0
  - HP CPU1 Peripheral bus writes in HP CPU1 region 0
  - HP CPU1 Peripheral bus reads in HP CPU1 region 1
  - HP CPU1 Peripheral bus writes in HP CPU1 region 1
- Monitoring SP bounds violations
  - HP CPU0 SP goes beyond the upper bound of the HP CPU0 SP monitored region
  - HP CPU0 SP goes below the lower bound of the HP CPU0 SP monitored region
  - HP CPU1 SP goes beyond the upper bound of the HP CPU1 SP monitored region
  - HP CPU1 SP goes below the lower bound of the HP CPU1 SP monitored region

The configuration process for region monitoring and SP monitoring is as follows:

1. Configure the monitored region and SP bounds.
   - Configure HP CPU0 Data bus region 0 with ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MIN_REG and ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MAX_REG.
   - Configure HP CPU0 Data bus region 1 with ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MIN_REG and ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MAX_REG.
   - Configure HP CPU0 Peripheral bus region 0 with ASSIST_DEBUG_CORE_0_AREA_PIF_0_MIN_REG and ASSIST_DEBUG_CORE_0_AREA_PIF_0_MAX_REG.
   - Configure HP CPU0 Peripheral bus region 1 with ASSIST_DEBUG_CORE_0_AREA_PIF_1_MIN_REG and ASSIST_DEBUG_CORE_0_AREA_PIF_1_MAX_REG.

- Configure HP CPU0 SP bounds with ASSIST_DEBUG_CORE_0_SP_MIN_REG and ASSIST_DEBUG_CORE_0_SP_MAX_REG.
- Configure HP CPU1 Data bus region 0 with ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MIN_REG and ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MAX_REG.
- Configure HP CPU1 Data bus region 1 with ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MIN_REG and ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MAX_REG.
- Configure HP CPU1 Peripheral bus region 0 with ASSIST_DEBUG_CORE_1_AREA_PIF_0_MIN_REG and ASSIST_DEBUG_CORE_1_AREA_PIF_0_MAX_REG.
- Configure HP CPU1 Peripheral bus region 1 with ASSIST_DEBUG_CORE_1_AREA_PIF_1_MIN_REG and ASSIST_DEBUG_CORE_1_AREA_PIF_1_MAX_REG.
- Configure HP CPU1 SP bounds with ASSIST_DEBUG_CORE_1_SP_MIN_REG and ASSIST_DEBUG_CORE_1_SP_MAX_REG.

2. Configure interrupts.
   - Configure ASSIST_DEBUG_CORE_0_INTR_ENA_REG or ASSIST_DEBUG_CORE_1_INTR_ENA_REG to enable the interrupt of a monitoring mode.
   - Read ASSIST_DEBUG_CORE_0_INTR_RAW_REG or ASSIST_DEBUG_CORE_1_INTR_RAW_REG to get the raw interrupt status of a monitoring mode.
   - Configure ASSIST_DEBUG_CORE_0_INTR_CLR_REG or ASSIST_DEBUG_CORE_1_INTR_CLR_REG to clear the interrupt of a monitoring mode.
3. Configure ASSIST_DEBUG_CORE_0_MONTR_ENA_REG or ASSIST_DEBUG_CORE_1_MONTR_ENA_REG to enable the monitoring mode(s). Various monitoring modes can be enabled at the same time.

Assuming that Debug Assistant module needs to monitor whether HP CPU0 Data bus has written to [A ~ B] address space, the user can enable monitoring in either HP CPU0 Data bus region 0 or region 1. The following configuration process is based on region 0:

1. Configure ASSIST_DEBUG_CORE_0_RCD_PDEBUGEN to 1 to enable HP CPU0 to update the PC signals to the Debug Assistant module.
2. Configure ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MIN_REG to Address A.
3. Configure ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MAX_REG to Address B.
4. Configure ASSIST_DEBUG_CORE_0_INTR_ENA_REG bit[1] to enable the interrupt for write operations by HP CPU0 Data bus in region 0.
5. Configure ASSIST_DEBUG_CORE_0_MONTR_ENA_REG bit[1] to enable monitoring write operations by HP CPU0 Data bus in region 0.
6. Configure interrupt matrix to map ASSIST_DEBUG_INTR into HP CPU0/1 interrupt (please refer to Chapter *12 Interrupt Matrix*).
7. After the interrupt is triggered:
   - Read ASSIST_DEBUG_CORE_0_INTR_RAW_REG and ASSIST_DEBUG_CORE_1_INTR_RAW_REG to learn which operation triggered the interrupt.

- If the interrupt is triggered by HP CPU0 region monitoring, read ASSIST_DEBUG_CORE_0_AREA_PC_REG for the HP CPU0 PC value, and ASSIST_DEBUG_CORE_0_AREA_SP_REG for the HP CPU0 SP.
- If the interrupt is triggered by SP monitoring, read ASSIST_DEBUG_CORE_0_SP_PC_REG for the HP CPU0 PC value.
- Write 1 to the corresponding bits of ASSIST_DEBUG_CORE_0_INTR_RAW_REG and ASSIST_DEBUG_CORE_1_INTR_CLR_REG to clear the interrupts.

21.5.2 PC Logging Configuration

Configure ASSIST_DEBUG_CORE_0_RCD_PDEBUGEN to 1 to enable HP CPU0 to update the PC signals to the Debug Assistant module. If ASSIST_DEBUG_CORE_0_RCD_RECORDEN is also configured to 1, ASSIST_DEBUG_CORE_0_RCD_PDEBUGPC_REG will record the HP CPU0's PC signal and ASSIST_DEBUG_CORE_0_RCD_PDEBUGSP_REG will record the HP CPU0 SP value. Otherwise, the two registers keep the original values.

When the HP CPU0 resets, ASSIST_DEBUG_CORE_0_RCD_EN_REG will reset, while ASSIST_DEBUG_CORE_0_RCD_PDEBUGPC_REG and ASSIST_DEBUG_CORE_0_RCD_PDEBUGSP_REG will not. Therefore, the two registers will keep the PC value and SP value at the HP CPU0 reset.

This is also the case for HP CPU1.

21.5.3 CPU/DMA Bus Access Logging Configuration

21.5.3.1 HP CPU0/1 Bus Access Logging Configuration

The configuration process for HP CPU0/1 bus access logging (SPM Monitor) is described below.

1. Configure monitored address space.
   - Configure SPM_MEM_MONITOR_LOG_MIN_REG and SPM_MEM_MONITOR_LOG_MAX_REG to specify monitored address space. The monitored address space should range from 0x3010_0000 to 0x3010_1FFF.
2. Configure the monitoring mode with SPM_MEM_MONITOR_LOG_MODE:
   - Write monitoring (whether the bus has write operations)
   - Word monitoring (whether the bus writes a specific word)
   - Halfword monitoring (whether the bus writes a specific halfword)
   - Byte monitoring (whether the bus writes a specific byte)
3. Configure the specific values to be monitored.
   - In word monitoring mode, SPM_MEM_MONITOR_LOG_CHECK_DATA_REG specifies the monitored word.
   - In halfword monitoring mode, SPM_MEM_MONITOR_LOG_CHECK_DATA_REG[15:0] specifies the monitored halfword.
   - In byte monitoring mode, SPM_MEM_MONITOR_LOG_CHECK_DATA_REG[7:0] specifies the monitored byte.

- SPM_MEM_MONITOR_LOG_DATA_MASK_REG is used to mask the byte specified in SPM_MEM_MONITOR_LOG_CHECK_DATA_REG. A masked byte can be any value. For example, in word monitoring, if SPM_MEM_MONITOR_LOG_CHECK_DATA_REG is configured to 0x01020304 and SPM_MEM_MONITOR_LOG_DATA_MASK_REG is configured to 0x1, then any writes of the data matching the 0x010203XX pattern by the bus will be recorded.

4. Configure the storage space for recorded data.
   - SPM_MEM_MONITOR_LOG_MEM_START_REG and SPM_MEM_MONITOR_LOG_MEM_END_REG specify the storage space for recorded data. The storage space must be in the range of 0x4080_0000 ~ 0x4087_FFFF.
   - Set SPM_MEM_MONITOR_LOG_MEM_ADDR_UPDATE_REG to update the value in SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG to SPM_MEM_MONITOR_LOG_MEM_START_REG.
   - Configure the permission for the SPM Monitor function of Debug Assistant module to access the HP L2MEM. Only when the access permission is enabled can the Debug Assistant module access the HP L2MEM. For more information, please refer to TEE_DMA_SPM_MON_PMS_W_REG and other relevant registers in Chapter 19 *Permission Control (PMS)*.
5. Configure the writing mode for the recorded data: loop mode or non-loop mode.
   - In loop mode, writing to the specified address space is performed in loops. When writing reaches the end address, it will return to the starting address and continue, overwriting the previously recorded data. Set SPM_MEM_MONITOR_LOG_MEM_LOOP_ENABLE to enable loop mode.
     For example, there are 10 write operations (1 ~ 10) to address space 0 ~ 4 during bus access. After the 5th operation writes to address 4, the 6th operation will start writing from address 0. The 6th to 10th operations will overwrite the previous data written by the 1th to 5th operations.
   - In non-loop mode, when writing reaches the end address, it will stop at the end address and dump the remaining data, not overwriting the previously recorded data. Clear SPM_MEM_MONITOR_LOG_MEM_LOOP_ENABLE to use non-loop mode.
     For example, there are 10 write operations (1 ~ 10) to address space 0 ~ 4 during bus access. After the 5th operation writes to address 4, the 6th to 10th write operations will stop at address 4 and will not be performed any more. Therefore, the address 0 ~ 4 stores the values written by the 1 ~ 5 operations and the values of the 6 ~ 10 operations are dumped.
6. Configure bus enable registers.
   - Enable HP CPU0 or HP CPU1 bus access logging with SPM_MEM_MONITOR_LOG_CORE_ENA. They can be enabled at the same time.

The Debug Assistant module first writes the recorded data of HP CPU0 or HP CPU1 to the internal buffer A, and then fetches the data from the buffer and writes it to the configured memory space. When the monitored behaviors are triggered continuously, the generated recording packets may fully occupy the buffer, making it unable to take any incoming packets. At this time, the module dumps these incoming packets and buffers a HP CPU LOST packet instead before the buffer reaches its capacity. However, it is only possible to know the bus type of the dumped packets when the HP CPU LOST packet was generated, but not the number of the dumped packets before the generation.

When bus access logging is finished, the recorded data can be read from memory for decoding. The recorded data is in three packet formats, namely HP CPU0 packet (corresponding to HP CPU0 Data bus), HP

CPU1 packet (corresponding to HP CPU1 Data bus), and HP CPU LOST packet. The packet formats are shown in Table 21.5-1 and 21.5-2.

Table 21.5-1. HP CPU0/1 Packet Format

Bit[63:33]	Bit[32]	Bit[31:3]	Bit[2:1]	Bit[0]
pc_offset	anchored (1)	addr_offset	format	anchored (0)

Table 21.5-2. HP CPU LOST Packet Format

Bit[31:7]	Bit[6:3]	Bit[2:1]	Bit[0]
reserved	lost_source	format	anchored (0)

It can be seen from the data packet formats that the size of the HP CPU0/1 packet is 64 bits and that of the HP CPU LOST packet is 32 bits. These packets contain the following fields:

- **format** – the packet type. 0: HP CPU0 packet; 1: HP CPU1 packet; 2: HP CPU LOST packet; 3: Reserved.
- **pc_offset** - the offset of the HP CPU0/1 PC register at the time of access. Actual PC = pc_offset + 0x0000_0000.
- **addr_offset** - the address offset of a write operation. Actual address = addr_offset + SPM_MEM_MONITOR_LOG_MIN_REG[31:4], 4'h0.
- **lost_source** - the dumped packets when the HP CPU LOST packet was generated.
  - Bit[4:3]: 0 indicates HP CPU0 packets were not dumped. 3 indicates HP CPU0 packets were dumped. Other values are reserved.
  - Bit[6:5]: 0 indicates HP CPU1 packets were not dumped. 3 indicates HP CPU1 packets were dumped. Other values are reserved.
- **anchored** - the location of the 32 bits in the data packet. 0: Lower 32 bits. 1: Higher 32 bits.

The internal buffer of the module is 32 bits wide. When the HP CPU0 and HP CPU1 bus access loggings are both enabled and the record data is generated at the same time, the HP CPU0 data packets are first buffered, then the HP CPU1 packets. The Debug Assistant will automatically fetch the buffered data and store it in 32-bit data width into the specified memory space.

In loop mode, data looping several times in the storage memory may cause residual data, which can interfere with packet parsing. For example, the lower 32 bits of a HP CPU0/1 packet are overwritten, thus making its higher 32 bits residual data. Therefore, users need to filter out the possible residual data in order to determine the starting position of the first valid packet with SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG. Once the starting position of the packet is identified, check the anchored bit value of the packet. If it is 0, the data will be retained. If it is 1, it will be dumped.

The process of packet parsing is described below:

- Determine whether there is a data overflow with SPM_MEM_MONITOR_LOG_MEM_FULL_FLAG.
  - If no, the address space to read is SPM_MEM_MONITOR_LOG_MEM_START_REG ~ SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG - 4.

  - If yes and the loop mode is enabled, the address space is SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG ~ SPM_MEM_MONITOR_LOG_MEM_END_REG and SPM_MEM_MONITOR_LOG_MEM_START_REG ~ SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG - 4.
  - If yes and loop mode is not enabled, the address space is SPM_MEM_MONITOR_LOG_MEM_START_REG ~ SPM_MEM_MONITOR_LOG_MEM_END_REG.
- Read and parse data from the starting address. Read 32 bits each time.

After packet parsing is completed, clear the SPM_MEM_MONITOR_LOG_MEM_FULL_FLAG flag bit by setting SPM_MEM_MONITOR_CLR_LOG_MEM_FULL_FLAG.

21.5.3.2 DMA Bus Access Logging Configuration

The Debug Assistant module supports a total of four DMA bus channel groups, DMA_0, DMA_1, DMA_2, and DMA_3 channel groups. DMA_0 and DMA_1 channel groups are reserved and cannot be used. The DMA bus for the DMA_2 channel group is the bus on which the AXI matrix accesses the HP L2MEM. The DMA bus for the DMA_3 channel group is the bus on which the AHB matrix accesses the HP L2MEM. For more information, see Chapter 7 *System and Memory* > Section 7.3.2. The configuration process for DMA bus access logging (L2MEM Monitor) is described below.

1. Configure monitored address space.
   - Configure L2_MEM_MONITOR_LOG_MIN_REG and L2_MEM_MONITOR_LOG_MAX_REG to specify monitored address space. The monitored address space should range from 0x4FF0_0000 to 0x4FFB_FFFF or from 0x8FF0_0000 to 0x8FFB_FFFF.
2. Configure the monitoring mode with L2_MEM_MONITOR_LOG_MODE:
   - Write monitoring (whether the bus has write operations)
   - Word monitoring (whether the bus writes a specific word)
   - Halfword monitoring (whether the bus writes a specific halfword)
   - Byte monitoring (whether the bus writes a specific byte)
3. Configure the specific values to be monitored.
   - In word monitoring mode, L2_MEM_MONITOR_LOG_CHECK_DATA_REG specifies the monitored word.
   - In halfword monitoring mode, L2_MEM_MONITOR_LOG_CHECK_DATA_REG[15:0] specifies the monitored halfword.
   - In byte monitoring mode, L2_MEM_MONITOR_LOG_CHECK_DATA_REG[7:0] specifies the monitored byte.
   - L2_MEM_MONITOR_LOG_DATA_MASK_REG is used to mask the byte specified in L2_MEM_MONITOR_LOG_CHECK_DATA_REG. A masked byte can be any value. For example, in word monitoring, if L2_MEM_MONITOR_LOG_CHECK_DATA_REG is configured to 0x01020304 and L2_MEM_MONITOR_LOG_DATA_MASK_REG is configured to 0x1, then any writes of the data matching the 0x010203XX pattern by the bus will be recorded.
4. Configure the storage space for recorded data.

- L2_MEM_MONITOR_LOG_MEM_START_REG and L2_MEM_MONITOR_LOG_MEM_END_REG specify the storage space for recorded data. The storage space must be in the range of 0x4080_0000 ~ 0x4087_FFFF.
- Set L2_MEM_MONITOR_LOG_MEM_ADDR_UPDATE_REG to update the value in L2_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG to L2_MEM_MONITOR_LOG_MEM_START_REG.
- Configure the permission for the L2MEM Monitor function of Debug Assistant module to access the HP L2MEM. Only when the access permission is enabled can the Debug Assistant module access the HP L2MEM. For more information, please refer to TEE_DMA_L2MEM_MON_PMS_W_REG and other relevant registers in Chapter 19 *Permission Control (PMS)*.

5. Configure the writing mode for the recorded data: loop mode or non-loop mode.
   - In loop mode, writing to the specified address space is performed in loops. When writing reaches the end address, it will return to the starting address and continue, overwriting the previously recorded data. Set L2_MEM_MONITOR_LOG_MEM_LOOP_ENABLE to enable loop mode.
     For example, there are 10 write operations (1 ~ 10) to address space 0 ~ 4 during bus access. After the 5th operation writes to address 4, the 6th operation will start writing from address 0. The 6th to 10th operations will overwrite the previous data written by the 1th to 5th operations.
   - In non-loop mode, when writing reaches the end address, it will stop at the end address and dump the remaining data, not overwriting the previously recorded data. Clear L2_MEM_MONITOR_LOG_MEM_LOOP_ENABLE to use non-loop mode.
     For example, there are 10 write operations (1 ~ 10) to address space 0 ~ 4 during bus access. After the 5th operation writes to address 4, the 6th to 10th write operations will stop at address 4 and will not be performed any more. Therefore, the address 0 ~ 4 stores the values written by the 1 ~ 5 operations and the values of the 6 ~ 10 operations are dumped.
   - See the example in 21.5.3.1 > step 5.
6. Configure bus enable registers.
   - Enable DMA_2 and DMA_3 channel groups bus access logging with L2_MEM_MONITOR_LOG_DMA_2_ENA and L2_MEM_MONITOR_LOG_DMA_3_ENA respectively. They can be enabled at the same time.

The Debug Assistant module first writes the DMA recorded data to the internal buffer B, and then fetches the data from the buffer and writes it to the configured memory space. When the monitored behaviors are triggered continuously, the generated recording packets may fully occupy the buffer, making it unable to take any incoming packets. At this time, the module dumps these incoming packets and buffers a DMA LOST packet instead before the buffer reaches its capacity. However, it is only possible to know the bus type of the dumped packets when the DMA LOST packet was generated, but not the number of the dumped packets before the generation.

When bus access logging is finished, the recorded data can be read from memory for decoding. The recorded data is in three packet formats, namely DMA_2 packet (corresponding to DMA_2 Data bus), DMA_3 packet (corresponding to DMA_3 Data bus), and DMA LOST packet. The packet formats are shown in Table 21.5-3, 21.5-4, and 21.5-5.

Table 21.5-3. DMA_2 Packet Format

Bit[31:9]	Bit[8:4]	Bit[3:1]	Bit[0]
addr_offset0	reserved	format	anchored (0)

Table 21.5-4. DMA_3 Packet Format

Bit[31:9]	Bit[8:4]	Bit[3:1]	Bit[0]
addr_offset1	reserved	format	anchored (0)

Table 21.5-5. DMA LOST Packet Format

Bit[31:10]	Bit[9:8]	Bit[7:4]	Bit[3:1]	Bit[0]
reserved	lost_source	reserved	format	anchored (0)

It can be seen from the data packet formats that the size of the DMA_2, DMA_3, and DMA LOST packet is 32 bits. These packets contain the following fields:

- **format** – the packet type. 4: DMA_2 packet; 5: DMA_3 packet; 6: DMA LOST packet; other values: Reserved.
- **addr_offset0** - the address offset of a write operation. Actual address = addr_offset0 + L2_MEM_MONITOR_LOG_MIN_REG[31:5],5'h0.
- **addr_offset1** - the address offset of a write operation. Actual address = addr_offset1 + L2_MEM_MONITOR_LOG_MIN_REG[31:2],2'h0.
- **lost_source** - the dumped packets when the DMA LOST packet was generated.
  - Bit[8]: 0 indicates DMA_2 packets were not dumped. 1 indicates DMA_2 packets were dumped.
  - Bit[9]: 0 indicates DMA_3 packets were not dumped. 1 indicates DMA_3 packets were dumped.
- **anchored** - the location of the 32 bits in the data packet. 0: Lower 32 bits. 1: Higher 32 bits.

The internal buffer of the module is 32 bits wide. When the bus access loggings of DMA_2 and DMA_3 channel groups are both enabled and the record data is generated at the same time, the DMA_2 data packets are first buffered, then the DMA_3 packets. The Debug Assistant will automatically fetch the buffered data and store it in 32-bit data width into the specified memory space.

The process of packet parsing is described below:

- Determine whether there is a data overflow with L2_MEM_MONITOR_LOG_MEM_FULL_FLAG.
  - If no, the address space to read is L2_MEM_MONITOR_LOG_MEM_START_REG ~ L2_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG - 4.
  - If yes and the loop mode is enabled, the address space is L2_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG ~ L2_MEM_MONITOR_LOG_MEM_END_REG and L2_MEM_MONITOR_LOG_MEM_START_REG ~ L2_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG - 4.
  - If yes and loop mode is not enabled, the address space is L2_MEM_MONITOR_LOG_MEM_START_REG ~ L2_MEM_MONITOR_LOG_MEM_END_REG.

- Read and parse data from the starting address. Read 32 bits each time.

After packet parsing is completed, clear the L2_MEM_MONITOR_LOG_MEM_FULL_FLAG flag bit by setting L2_MEM_MONITOR_CLR_LOG_MEM_FULL_FLAG.

21.6 Register Summary

The addresses of **HP CPU bus logging configuration registers** (see 21.6.1) in this section are relative to the **SPM Monitor** base address. The addresses of **DMA bus logging configuration registers** (see 21.6.2) in this section are relative to the **L2MEM Monitor** base address. The addresses of other registers (see 21.6.3) are relative to the **Bus Monitor** base address. All base addresses are provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

21.6.1 HP CPU Bus Logging Configuration Register Summary

Name	Description	Address	Access
HP CPU bus logging configuration registers			
SPM_MEM_MONITOR_LOG_SETTING_REG	Configures bus access logging	0x0000	R/W
SPM_MEM_MONITOR_LOG_CHECK_DATA_REG	Configures monitored data in bus access logging	0x0008	R/W
SPM_MEM_MONITOR_LOG_DATA_MASK_REG	Configures masked data in bus access logging	0x000C	R/W
SPM_MEM_MONITOR_LOG_MIN_REG	Configures monitored address space in bus access logging	0x0010	R/W
SPM_MEM_MONITOR_LOG_MAX_REG	Configures monitored address space in bus access logging	0x0014	R/W
SPM_MEM_MONITOR_LOG_MEM_START_REG	Configures the starting address of the memory for recorded data	0x0018	R/W
SPM_MEM_MONITOR_LOG_MEM_END_REG	Configures the end address of the memory for recorded data	0x001C	R/W
SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG	Represents the address for the next write	0x0020	RO
SPM_MEM_MONITOR_LOG_MEM_ADDR_UPDATE_REG	Updates the address for the next write to the starting address for the recorded data	0x0024	WT
SPM_MEM_MONITOR_LOG_MEM_FULL_FLAG_REG	Represents the logging overflow status	0x0028	varies
Clock control register			
SPM_MEM_MONITOR_CLOCK_GATE_REG	Clock control register	0x002C	R/W
Version control register			
SPM_MEM_MONITOR_DATE_REG	Version control register	0x03FC	R/W

21.6.2 DMA Bus Logging Configuration Register Summary

Name	Description	Address	Access
DMA bus logging configuration registers			

Name	Description	Address	Access
L2_MEM_MONITOR_LOG_SETTING_REG	Configures bus access logging	0x0000	R/W
L2_MEM_MONITOR_LOG_SETTING1_REG	Configures bus access logging	0x0004	R/W
L2_MEM_MONITOR_LOG_CHECK_DATA_REG	Configures monitored data in bus access logging	0x0008	R/W
L2_MEM_MONITOR_LOG_DATA_MASK_REG	Configures masked data in bus access logging	0x000C	R/W
L2_MEM_MONITOR_LOG_MIN_REG	Configures monitored address space in bus access logging	0x0010	R/W
L2_MEM_MONITOR_LOG_MAX_REG	Configures monitored address space in bus access logging	0x0014	R/W
L2_MEM_MONITOR_LOG_MEM_START_REG	Configures the starting address of the memory for recorded data	0x0018	R/W
L2_MEM_MONITOR_LOG_MEM_END_REG	Configures the end address of the memory for recorded data	0x001C	R/W
L2_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG	Represents the address for the next write	0x0020	RO
L2_MEM_MONITOR_LOG_MEM_ADDR_UPDATE_REG	Updates the address for the next write to the starting address for the recorded data	0x0024	WT
L2_MEM_MONITOR_LOG_MEM_FULL_FLAG_REG	Represents the logging overflow status	0x0028	varies
Clock control register			
L2_MEM_MONITOR_CLOCK_GATE_REG	Clock control register	0x002C	R/W
Version control register			
L2_MEM_MONITOR_DATE_REG	Version control register	0x03FC	R/W

21.6.3 Summary of Other Registers

Name	Description	Address	Access
Monitor configuration registers			
ASSIST_DEBUG_CORE_0_MONTR_ENA_REG	Configures whether to enable HP CPU0 monitoring	0x0000	R/W
ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MIN_REG	Configures the lower bound address of region 0 monitored on HP CPU0 Data bus	0x0010	R/W
ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MAX_REG	Configures the upper bound address of region 0 monitored on HP CPU0 Data bus	0x0014	R/W
ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MIN_REG	Configures the lower bound address of region 1 monitored on HP CPU0 Data bus	0x0018	R/W

Name	Description	Address	Access
ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MAX_REG	Configures the upper bound address of region 1 monitored on HP CPU0 Data bus	0x001C	R/W
ASSIST_DEBUG_CORE_0_AREA_PIF_0_MIN_REG	Configures the lower bound address of region 0 monitored on HP CPU0 Peripheral bus	0x0020	R/W
ASSIST_DEBUG_CORE_0_AREA_PIF_0_MAX_REG	Configures the upper bound address of region 0 monitored on HP CPU0 Peripheral bus	0x0024	R/W
ASSIST_DEBUG_CORE_0_AREA_PIF_1_MIN_REG	Configures the lower bound address of region 1 monitored on HP CPU0 Peripheral bus	0x0028	R/W
ASSIST_DEBUG_CORE_0_AREA_PIF_1_MAX_REG	Configures the upper bound address of region 1 monitored on HP CPU0 Peripheral bus	0x002C	R/W
ASSIST_DEBUG_CORE_0_AREA_PC_REG	Represents the PC status under HP CPU0 region monitoring	0x0030	RO
ASSIST_DEBUG_CORE_0_AREA_SP_REG	Represents the SP status under HP CPU0 region monitoring	0x0034	RO
ASSIST_DEBUG_CORE_0_SP_MIN_REG	Configures the lower bound address of the HP CPU0 SP monitored region	0x0038	R/W
ASSIST_DEBUG_CORE_0_SP_MAX_REG	Configures the upper bound address of the HP CPU0 SP monitored region	0x003C	R/W
ASSIST_DEBUG_CORE_0_SP_PC_REG	Represents the PC status under HP CPU0 SP monitoring	0x0040	RO
ASSIST_DEBUG_CORE_1_MONTR_ENA_REG	Configures whether to enable HP CPU1 monitoring	0x0080	R/W
ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MIN_REG	Configures the lower bound address of region 0 monitored on HP CPU1 Data bus	0x0090	R/W
ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MAX_REG	Configures the upper bound address of region 0 monitored on HP CPU1 Data bus	0x0094	R/W
ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MIN_REG	Configures the lower bound address of region 1 monitored on HP CPU1 Data bus	0x0098	R/W
ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MAX_REG	Configures the upper bound address of region 1 monitored on HP CPU1 Data bus	0x009C	R/W
ASSIST_DEBUG_CORE_1_AREA_PIF_0_MIN_REG	Configures the lower bound address of region 0 monitored on HP CPU1 Peripheral bus	0x00A0	R/W

Name	Description	Address	Access
ASSIST_DEBUG_CORE_1_AREA_PIF_0_MAX_REG	Configures the upper bound address of region 0 monitored on HP CPU1 Peripheral bus	0x00A4	R/W
ASSIST_DEBUG_CORE_1_AREA_PIF_1_MIN_REG	Configures the lower bound address of region 1 monitored on HP CPU1 Peripheral bus	0x00A8	R/W
ASSIST_DEBUG_CORE_1_AREA_PIF_1_MAX_REG	Configures the upper bound address of region 1 monitored on HP CPU1 Peripheral bus	0x00AC	R/W
ASSIST_DEBUG_CORE_1_AREA_PC_REG	Represents the PC status under HP CPU1 region monitoring	0x00B0	RO
ASSIST_DEBUG_CORE_1_AREA_SP_REG	Represents the SP status under HP CPU1 region monitoring	0x00B4	RO
ASSIST_DEBUG_CORE_1_SP_MIN_REG	Configures the lower bound address of the HP CPU1 SP monitored region	0x00B8	R/W
ASSIST_DEBUG_CORE_1_SP_MAX_REG	Configures the upper bound address of the HP CPU1 SP monitored region	0x00BC	R/W
ASSIST_DEBUG_CORE_1_SP_PC_REG	Represents the PC status under HP CPU1 SP monitoring	0x00C0	RO
Interrupt configuration registers			
ASSIST_DEBUG_CORE_0_INTR_RAW_REG	HP CPU0 raw interrupt status register	0x0004	RO
ASSIST_DEBUG_CORE_0_INTR_ENA_REG	HP CPU0 interrupt enable register	0x0008	R/W
ASSIST_DEBUG_CORE_0_INTR_CLR_REG	HP CPU0 interrupt clear register	0x000C	WT
ASSIST_DEBUG_CORE_1_INTR_RAW_REG	HP CPU1 raw interrupt status register	0x0084	RO
ASSIST_DEBUG_CORE_1_INTR_ENA_REG	HP CPU1 interrupt enable register	0x0088	R/W
ASSIST_DEBUG_CORE_1_INTR_CLR_REG	HP CPU1 interrupt clear register	0x008C	WT
PC logging configuration register			
ASSIST_DEBUG_CORE_0_RCD_EN_REG	HP CPU0 PC logging enable register	0x0044	R/W
ASSIST_DEBUG_CORE_1_RCD_EN_REG	HP CPU1 PC logging enable register	0x00C4	R/W
PC logging status registers			
ASSIST_DEBUG_CORE_0_RCD_PDEBUGPC_REG	HP CPU0 PC logging register	0x0048	RO
ASSIST_DEBUG_CORE_0_RCD_PDEBUGSP_REG	HP CPU0 SP logging register	0x004C	RO
ASSIST_DEBUG_CORE_1_RCD_PDEBUGPC_REG	HP CPU1 PC logging register	0x00C8	RO
ASSIST_DEBUG_CORE_1_RCD_PDEBUGSP_REG	HP CPU1 SP logging register	0x00CC	RO
CPU status registers			
ASSIST_DEBUG_CORE_0_LASTPC_BEFORE_EXCEPTION_REG	PC of the last command before HP CPU0 enters exception	0x0070	RO

Name	Description	Address	Access
ASSIST_DEBUG_CORE_0_DEBUG_MODE_REG	HP CPU0 debug mode status register	0x0074	RO
ASSIST_DEBUG_CORE_1_LASTPC_BEFORE_EXCEPTION_REG	PC of the last command before HP CPU1 enters exception	0x00F0	RO
ASSIST_DEBUG_CORE_1_DEBUG_MODE_REG	HP CPU1 debug mode status register	0x00F4	RO
Clock control register			
ASSIST_DEBUG_CLOCK_GATE_REG	Clock control register	0x0108	R/W
Version control register			
ASSIST_DEBUG_DATE_REG	Version control register	0x03FC	R/W

21.7 Registers

The addresses of **HP CPU bus logging configuration registers** (see 21.6.1) in this section are relative to the **SPM Monitor** base address. The addresses of **DMA bus logging configuration registers** (see 21.6.2) in this section are relative to the **L2MEM Monitor** base address. The addresses of other registers (see 21.7.3) are relative to the **Bus Monitor** base address. All base addresses are provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

21.7.1 HP CPU Bus Logging Configuration Registers

Register 21.1. SPM_MEM_MONITOR_LOG_SETTING_REG (0x0000)

	(reserved)	SPM_MEM_MONITOR_LOG_CORE_ENA	(reserved)	SPM_MEM_MONITOR_LOG_MEM_LOOP_ENABLE	SPM_MEM_MONITOR_LOG_MODE
Bits	31–10	9–8	7–5	4	3–0
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	1	0

SPM_MEM_MONITOR_LOG_MODE Configures monitoring modes.

bit[0]: Configures write monitoring.
0: Disable
1: Enable
bit[1]: Configures word monitoring.
0: Disable
1: Enable
bit[2]: Configures halfword monitoring.
0: Disable
1: Enable
bit[3]: Configures byte monitoring.
0: Disable
1: Enable
(R/W)

SPM_MEM_MONITOR_LOG_MEM_LOOP_ENABLE Configures the writing mode for recorded data.

1: Loop mode
0: Non-loop mode
(R/W)

Continued on the next page...

Register 21.1. SPM_MEM_MONITOR_LOG_SETTING_REG (0x0000)

Continued from the previous page...

SPM_MEM_MONITOR_LOG_CORE_ENA Configures whether to enable HP CPU0 or HP CPU1 bus access logging.
bit[0]: Configures whether to enable HP CPU0 bus access logging.
0: Disable
1: Enable
bit[1]: Configures whether to enable HP CPU1 bus access logging.
0: Disable
1: Enable
(R/W)

Register 21.2. SPM_MEM_MONITOR_LOG_CHECK_DATA_REG (0x0008)

SPM_MEM_MONITOR_LOG_CHECK_DATA	
31 ... 0	
0	Reset

SPM_MEM_MONITOR_LOG_CHECK_DATA Configures the data to be monitored during bus accessing. (R/W)

Register 21.3. SPM_MEM_MONITOR_LOG_DATA_MASK_REG (0x000C)

(reserved)	SPM_MEM_MONITOR_LOG_DATA_MASK	
31 ... 4	3 ... 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

SPM_MEM_MONITOR_LOG_DATA_MASK Configures which byte(s) in SPM_MEM_MONITOR_LOG_CHECK_DATA_REG to mask.

bit[0]: Configures whether to mask the least significant byte of SPM_MEM_MONITOR_LOG_CHECK_DATA_REG.

0: Not mask

1: Mask

bit[1]: Configures whether to mask the second least significant byte of SPM_MEM_MONITOR_LOG_CHECK_DATA_REG.

0: Not mask

1: Mask

bit[2]: Configures whether to mask the second most significant byte of SPM_MEM_MONITOR_LOG_CHECK_DATA_REG.

0: Not mask

1: Mask

bit[3]: Configures whether to mask the most significant byte of SPM_MEM_MONITOR_LOG_CHECK_DATA_REG.

0: Not mask

1: Mask

(R/W)

Register 21.4. SPM_MEM_MONITOR_LOG_MIN_REG (0x0010)

SPM_MEM_MONITOR_LOG_MIN	
31 ... 0	
0	Reset

SPM_MEM_MONITOR_LOG_MIN Configures the lower bound address of the monitored address space. (R/W)

Register 21.5. SPM_MEM_MONITOR_LOG_MAX_REG (0x0014)

SPM_MEM_MONITOR_LOG_MAX	
31 ... 0	
0	Reset

SPM_MEM_MONITOR_LOG_MAX Configures the upper bound address of the monitored address space. (R/W)

Register 21.6. SPM_MEM_MONITOR_LOG_MEM_START_REG (0x0018)

SPM_MEM_MONITOR_LOG_MEM_START	
31 ... 0	
0	Reset

SPM_MEM_MONITOR_LOG_MEM_START Configures the starting address of the storage space for recorded data. (R/W)

Register 21.7. SPM_MEM_MONITOR_LOG_MEM_END_REG (0x001C)

SPM_MEM_MONITOR_LOG_MEM_END	
31 ... 0	
0	Reset

SPM_MEM_MONITOR_LOG_MEM_END Configures the ending address of the storage space for recorded data. (R/W)

Register 21.8. SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG (0x0020)

SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR	
31 … 0	
0	Reset

SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR Represents the address of the next write. (RO)

Register 21.9. SPM_MEM_MONITOR_LOG_MEM_ADDR_UPDATE_REG (0x0024)

(reserved)	SPM_MEM_MONITOR_LOG_MEM_ADDR_UPDATE	
31 … 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

SPM_MEM_MONITOR_LOG_MEM_ADDR_UPDATE Configures whether to update the value in SPM_MEM_MONITOR_LOG_MEM_START_REG to SPM_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG.

1: Update

0: Not update

(WT)

Register 21.10. SPM_MEM_MONITOR_LOG_MEM_FULL_FLAG_REG (0x0028)

(reserved)	SPM_MEM_MONITOR_CLR_LOG_MEM_FULL_FLAG	SPM_MEM_MONITOR_LOG_MEM_FULL_FLAG	
31 ... 2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	Reset

SPM_MEM_MONITOR_LOG_MEM_FULL_FLAG Represents whether data overflows the storage space.

0: Not overflow

1: Overflow

(RO)

SPM_MEM_MONITOR_CLR_LOG_MEM_FULL_FLAG Configures whether to clear the SPM_MEM_MONITOR_LOG_MEM_FULL_FLAG flag bit.

0: Not clear (default)

1: Clear

(WT)

Register 21.11. SPMMEM_MONITOR_CLOCK_GATE_REG (0x002C)

(reserved)	SPM_MEM_MONITOR_CLK_EN	
31 ... 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	1	Reset

SPM_MEM_MONITOR_CLK_EN Configures whether to enable the register clock gating.

0: Disable

1: Enable

(R/W)

Register 21.12. SPM_MEM_MONITOR_DATE_REG (0x03FC)

(reserved)	SPM_MEM_MONITOR_DATE	
31 28	27 0	
0 0 0 0	0x2302220	Reset

SPM_MEM_MONITOR_DATE Version control register. (R/W)

21.7.2 DMA Bus Logging Configuration Registers

Register 21.13. L2_MEM_MONITOR_LOG_SETTING_REG (0x0000)

	(reserved)	L2_MEM_MONITOR_LOG_MEM_LOOP_ENABLE	L2_MEM_MONITOR_LOG_MODE
Bits	31 … 5	4	3 … 0
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	1	0

L2_MEM_MONITOR_LOG_MODE Configures monitoring modes.

bit[0]: Configures write monitoring.

0: Disable

1: Enable

bit[1]: Configures word monitoring.

0: Disable

1: Enable

bit[2]: Configures halfword monitoring.

0: Disable

1: Enable

bit[3]: Configures byte monitoring.

0: Disable

1: Enable

(R/W)

L2_MEM_MONITOR_LOG_MEM_LOOP_ENABLE Configures the writing mode for recorded data.

1: Loop mode

0: Non-loop mode

(R/W)

Register 21.14. L2_MEM_MONITOR_LOG_SETTING1_REG (0x0004)

(reserved)	L2_MEM_MONITOR_LOG_DMA_3_ENA	(reserved)	L2_MEM_MONITOR_LOG_DMA_2_ENA	
31 … 9	8	7 … 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	Reset

L2_MEM_MONITOR_LOG_DMA_2_ENA Configures whether to enable DMA_2 bus access logging.
 0: Disable
 1: Enable
 (R/W)

L2_MEM_MONITOR_LOG_DMA_3_ENA Configures whether to enable DMA_3 bus access logging.
 0: Disable
 1: Enable
 (R/W)

Register 21.15. L2_MEM_MONITOR_LOG_CHECK_DATA_REG (0x0008)

L2_MEM_MONITOR_LOG_CHECK_DATA	
31 … 0	
0	Reset

L2_MEM_MONITOR_LOG_CHECK_DATA Configures the data to be monitored during bus accessing. (R/W)

Register 21.16. L2_MEM_MONITOR_LOG_DATA_MASK_REG (0x000C)

	(reserved)	L2_MEM_MONITOR_LOG_DATA_MASK	
Bits	31 … 4	3 … 0	
	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

L2_MEM_MONITOR_LOG_DATA_MASK Configures which byte(s) in L2_MEM_MONITOR_LOG_CHECK_DATA_REG to mask.

bit[0]: Configures whether to mask the least significant byte of L2_MEM_MONITOR_LOG_CHECK_DATA_REG.
0: Not mask
1: Mask
bit[1]: Configures whether to mask the second least significant byte of L2_MEM_MONITOR_LOG_CHECK_DATA_REG.
0: Not mask
1: Mask
bit[2]: Configures whether to mask the second most significant byte of L2_MEM_MONITOR_LOG_CHECK_DATA_REG.
0: Not mask
1: Mask
bit[3]: Configures whether to mask the most significant byte of L2_MEM_MONITOR_LOG_CHECK_DATA_REG.
0: Not mask
1: Mask
(R/W)

Register 21.17. L2_MEM_MONITOR_LOG_MIN_REG (0x0010)

	L2_MEM_MONITOR_LOG_MIN	
Bits	31 … 0	
	0	Reset

L2_MEM_MONITOR_LOG_MIN Configures the lower bound address of the monitored address space. (R/W)

Register 21.18. L2_MEM_MONITOR_LOG_MAX_REG (0x0014)

L2_MEM_MONITOR_LOG_MAX	
31 ... 0	
0	Reset

L2_MEM_MONITOR_LOG_MAX Configures the upper bound address of the monitored address space. (R/W)

Register 21.19. L2_MEM_MONITOR_LOG_MEM_START_REG (0x0018)

L2_MEM_MONITOR_LOG_MEM_START	
31 ... 0	
0	Reset

L2_MEM_MONITOR_LOG_MEM_START Configures the starting address of the storage space for recorded data. (R/W)

Register 21.20. L2_MEM_MONITOR_LOG_MEM_END_REG (0x001C)

L2_MEM_MONITOR_LOG_MEM_END	
31 ... 0	
0	Reset

L2_MEM_MONITOR_LOG_MEM_END Configures the ending address of the storage space for recorded data. (R/W)

Register 21.21. L2_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG (0x0020)

L2_MEM_MONITOR_LOG_MEM_CURRENT_ADDR	
31 … 0	
0	Reset

L2_MEM_MONITOR_LOG_MEM_CURRENT_ADDR Represents the address of the next write. (RO)

Register 21.22. L2_MEM_MONITOR_LOG_MEM_ADDR_UPDATE_REG (0x0024)

(reserved)	L2_MEM_MONITOR_LOG_MEM_ADDR_UPDATE	
31 … 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

L2_MEM_MONITOR_LOG_MEM_ADDR_UPDATE Configures whether to update the value in L2_MEM_MONITOR_LOG_MEM_START_REG to L2_MEM_MONITOR_LOG_MEM_CURRENT_ADDR_REG.

1: Update

0: Not update (default)

(WT)

Register 21.23. L2_MEM_MONITOR_LOG_MEM_FULL_FLAG_REG (0x0028)

(reserved)	L2_MEM_MONITOR_CLR_LOG_MEM_FULL_FLAG	L2_MEM_MONITOR_LOG_MEM_FULL_FLAG	
31 ... 2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	Reset

L2_MEM_MONITOR_LOG_MEM_FULL_FLAG Represents whether data overflows the storage space.

0: Not overflow

1: Overflow

(RO)

L2_MEM_MONITOR_CLR_LOG_MEM_FULL_FLAG Configures whether to clear the L2_MEM_MONITOR_LOG_MEM_FULL_FLAG flag bit.

0: Not clear (default)

1: Clear

(WT)

Register 21.24. L2_MEM_MONITOR_CLOCK_GATE_REG (0x002C)

(reserved)	L2_MEM_MONITOR_CLK_EN	
31 ... 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	1	Reset

L2_MEM_MONITOR_CLK_EN Configures whether to enable the register clock gating.

0: Disable

1: Enable

(R/W)

Register 21.25. L2_MEM_MONITOR_DATE_REG (0x03FC)

(reserved)	L2_MEM_MONITOR_DATE	
31 28	27 0	
0 0 0 0	0x2302220	Reset

L2_MEM_MONITOR_DATE Version control register. (R/W)

21.7.3 Other Registers

Register 21.26. ASSIST_DEBUG_CORE_0_MONTR_ENA_REG (0x0000)

Bits	31–10	9	8	7	6	5	4	3	2	1	0	
Field	(reserved)	ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_ENA	ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_ENA	ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_ENA	ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_ENA	ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_ENA	ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_ENA	ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_ENA	ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_ENA	ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_ENA	ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_ENA	
Value	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	0	Reset

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_ENA Configures whether to monitor read operations in HP CPU0 region 0 by the HP CPU0 Data bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_ENA Configures whether to monitor write operations in HP CPU0 region 0 by the HP CPU0 Data bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_ENA Configures whether to monitor read operations in HP CPU0 region 1 by the HP CPU0 Data bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_ENA Configures whether to monitor write operations in HP CPU0 region 1 by the HP CPU0 Data bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_ENA Configures whether to monitor read operations in region 0 by the Peripheral bus.
0: Not monitor
1: Monitor
(R/W)

Continued on the next page...

Register 21.26. ASSIST_DEBUG_CORE_0_MONTR_ENA_REG (0x0000)

Continued from the previous page...

ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_ENA Configures whether to monitor write operations in HP CPU0 region 0 by the HP CPU0 Peripheral bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_ENA Configures whether to monitor read operations in HP CPU0 region 1 by the HP CPU0 Peripheral bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_ENA Configures whether to monitor write operations in HP CPU0 region 1 by the HP CPU0 Peripheral bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_ENA Configures whether to monitor HP CPU0 SP going below the lower bound address of HP CPU0 SP monitored region.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_ENA Configures whether to monitor HP CPU0 SP going beyond the upper bound address of HP CPU0 SP monitored region.
0: Not monitor
1: Monitor
(R/W)

Register 21.27. ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MIN_REG (0x0010)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MIN	
31 … 0	
0xffffffff	Reset

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MIN Configures the lower bound address of HP CPU0 Data bus region 0. (R/W)

Register 21.28. ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MAX_REG (0x0014)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MAX	
31 … 0	
0	Reset

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_MAX Configures the upper bound address of HP CPU0 Data bus region 0. (R/W)

Register 21.29. ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MIN_REG (0x0018)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MIN	
31 ... 0	
0xffffffff	Reset

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MIN Configures the lower bound address of HP CPU0 Data bus region 1. (R/W)

Register 21.30. ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MAX_REG (0x001C)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MAX	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_MAX Configures the upper bound address of HP CPU0 Data bus region 1. (R/W)

Register 21.31. ASSIST_DEBUG_CORE_0_AREA_PIF_0_MIN_REG (0x0020)

ASSIST_DEBUG_CORE_0_AREA_PIF_0_MIN (31 … 0)	
0xffffffff	Reset

ASSIST_DEBUG_CORE_0_AREA_PIF_0_MIN Configures the lower bound address of HP CPU0 Peripheral bus region 0. (R/W)

Register 21.32. ASSIST_DEBUG_CORE_0_AREA_PIF_0_MAX_REG (0x0024)

ASSIST_DEBUG_CORE_0_AREA_PIF_0_MAX (31 … 0)	
0	Reset

ASSIST_DEBUG_CORE_0_AREA_PIF_0_MAX Configures the upper bound address of HP CPU0 Peripheral bus region 0. (R/W)

Register 21.33. ASSIST_DEBUG_CORE_0_AREA_PIF_1_MIN_REG (0x0028)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_MIN	
31 0	
0xffffffff	Reset

ASSIST_DEBUG_CORE_0_AREA_PIF_1_MIN Configures the lower bound address of HP CPU0 Peripheral bus region 1. (R/W)

Register 21.34. ASSIST_DEBUG_CORE_0_AREA_PIF_1_MAX_REG (0x002C)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_MAX	
31 0	
0	Reset

ASSIST_DEBUG_CORE_0_AREA_PIF_1_MAX Configures the upper bound address of HP CPU0 Peripheral bus region 1. (R/W)

Register 21.35. ASSIST_DEBUG_CORE_0_AREA_PC_REG (0x0030)

ASSIST_DEBUG_CORE_0_AREA_PC	
31 0	
0	Reset

ASSIST_DEBUG_CORE_0_AREA_PC Represents the HP CPU0 PC value when an interrupt is triggered during region monitoring. (RO)

Register 21.36. ASSIST_DEBUG_CORE_0_AREA_SP_REG (0x0034)

ASSIST_DEBUG_CORE_0_AREA_SP	
31 … 0	
0	Reset

ASSIST_DEBUG_CORE_0_AREA_SP Represents the HP CPU0 SP value when an interrupt is triggered during region monitoring. (RO)

Register 21.37. ASSIST_DEBUG_CORE_0_SP_MIN_REG (0x0038)

ASSIST_DEBUG_CORE_0_SP_MIN	
31 … 0	
0	Reset

ASSIST_DEBUG_CORE_0_SP_MIN Configures the lower bound address of the HP CPU0 SP monitored region. (R/W)

Register 21.38. ASSIST_DEBUG_CORE_0_SP_MAX_REG (0x003C)

ASSIST_DEBUG_CORE_0_SP_MAX	
31 … 0	
0xffffffff	Reset

ASSIST_DEBUG_CORE_0_SP_MAX Configures the upper bound address of the HP CPU0 SP monitored region. (R/W)

Register 21.39. ASSIST_DEBUG_CORE_0_SP_PC_REG (0x0040)

ASSIST_DEBUG_CORE_0_SP_PC	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_0_SP_PC Represents the HP CPU0 PC value during SP monitoring. (RO)

Register 21.40. ASSIST_DEBUG_CORE_0_INTR_RAW_REG (0x0004)

Field	(reserved)	ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_RAW	ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_RAW	ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_RAW	ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_RAW	ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_RAW	ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_RAW	ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_RAW	ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_RAW	ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_RAW	ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_RAW
Bits	31 … 10	9	8	7	6	5	4	3	2	1	0
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	0

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_INT. (RO)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_INT. (RO)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_INT. (RO)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_INT. (RO)

ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_INT. (RO)

ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_INT. (RO)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_INT. (RO)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_INT. (RO)

ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_INT. (RO)

ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_RAW The raw interrupt status of ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_INT. (RO)

Register 21.41. ASSIST_DEBUG_CORE_0_INTR_ENA_REG (0x0008)

Bits	31–10	9	8	7	6	5	4	3	2	1	0
Field	(reserved)	ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_INTR_ENA	ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_INTR_ENA	ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_INTR_ENA	ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_INTR_ENA	ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_INTR_ENA	ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_INTR_ENA	ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_INTR_ENA	ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_INTR_ENA	ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_INTR_ENA	ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_INTR_ENA
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	0

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_INT. (R/W)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_INT. (R/W)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_INT. (R/W)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_INT. (R/W)

ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_INT. (R/W)

ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_INT. (R/W)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_INT. (R/W)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_INT. (R/W)

ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_INT. (R/W)

ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_INT. (R/W)

Register 21.42. ASSIST_DEBUG_CORE_0_INTR_CLR_REG (0x000C)

(reserved)	ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_CLR	(ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_CLR	(ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_CLR	(ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_CLR	(ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_CLR	(ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_CLR	(ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_CLR	(ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_CLR	(ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_CLR	(ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_CLR	
31 10	9	8	7	6	5	4	3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	0	Reset

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_RD_INT. (WT)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_AREA_DRAM0_0_WR_INT. (WT)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_RD_INT. (WT)

ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_AREA_DRAM0_1_WR_INT. (WT)

ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_AREA_PIF_0_RD_INT. (WT)

ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_AREA_PIF_0_WR_INT. (WT)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_AREA_PIF_1_RD_INT. (WT)

ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_AREA_PIF_1_WR_INT. (WT)

ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_SP_SPILL_MIN_INT. (WT)

ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_CLR Write 1 to clear ASSIST_DEBUG_CORE_0_SP_SPILL_MAX_INT. (WT)

Register 21.43. ASSIST_DEBUG_CORE_0_RCD_EN_REG (0x0044)

(reserved)	ASSIST_DEBUG_CORE_0_RCD_PDEBUGEN	ASSIST_DEBUG_CORE_0_RCD_RECORDEN	
31 … 2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	Reset

ASSIST_DEBUG_CORE_0_RCD_RECORDEN Configures whether to enable HP CPU0 PC logging.
0: Disable
1: ASSIST_DEBUG_CORE_0_RCD_PDEBUGPC_REG starts to record HP CPU0 PC in real time
(R/W)

ASSIST_DEBUG_CORE_0_RCD_PDEBUGEN Configures whether to enable HP CPU0 debugging.
0: Disable
1: HP CPU0 outputs HP CPU0 PC
(R/W)

Register 21.44. ASSIST_DEBUG_CORE_0_RCD_PDEBUGPC_REG (0x0048)

ASSIST_DEBUG_CORE_0_RCD_PDEBUGPC	
31 … 0	
0x000000	Reset

ASSIST_DEBUG_CORE_0_RCD_PDEBUGPC Represents the PC value at HP CPU0 reset. (RO)

Register 21.45. ASSIST_DEBUG_CORE_0_RCD_PDEBUGSP_REG (0x004C)

ASSIST_DEBUG_CORE_0_RCD_PDEBUGSP	
31 … 0	
0x000000	Reset

ASSIST_DEBUG_CORE_0_RCD_PDEBUGSP Represents the HP CPU0 SP. (RO)

Register 21.46. ASSIST_DEBUG_CORE_0_LASTPC_BEFORE_EXCEPTION_REG (0x0070)

ASSIST_DEBUG_CORE_0_LASTPC_BEFORE_EXC	
31 … 0	
0	Reset

ASSIST_DEBUG_CORE_0_LASTPC_BEFORE_EXC Represents the HP CPU0 PC of the last command before the HP CPU0 enters exception. (RO)

Register 21.47. ASSIST_DEBUG_CORE_0_DEBUG_MODE_REG (0x0074)

Bits	31 … 2	1	0	
Field	(reserved)	ASSIST_DEBUG_CORE_0_DEBUG_MODULE_ACTIVE	ASSIST_DEBUG_CORE_0_DEBUG_MODE	
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	Reset

ASSIST_DEBUG_CORE_0_DEBUG_MODE Represents whether HP CPU0 is in debugging mode.

1: In debugging mode

0: Not in debugging mode

(RO)

ASSIST_DEBUG_CORE_0_DEBUG_MODULE_ACTIVE Represents the status of the HP CPU0 debug module.

1: Active status

Other: Inactive status

(RO)

Register 21.48. ASSIST_DEBUG_CORE_1_MONTR_ENA_REG (0x0080)

Bits	31–10	9	8	7	6	5	4	3	2	1	0	
Field	(reserved)	ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_ENA	ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_ENA	ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_ENA	ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_ENA	ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_ENA	ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_ENA	ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_ENA	ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_ENA	ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_ENA	ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_ENA	
Value	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	0	Reset

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_ENA Configures whether to monitor read operations in HP CPU1 region 0 by the HP CPU1 Data bus.

0: Not monitor

1: Monitor

(R/W)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_ENA Configures whether to monitor write operations in HP CPU1 region 0 by the HP CPU1 Data bus.

0: Not monitor

1: Monitor

(R/W)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_ENA Configures whether to monitor read operations in HP CPU1 region 1 by the HP CPU1 Data bus.

0: Not monitor

1: Monitor

(R/W)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_ENA Configures whether to monitor write operations in HP CPU1 region 1 by the HP CPU1 Data bus.

0: Not monitor

1: Monitor

(R/W)

ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_ENA Configures whether to monitor read operations in HP CPU1 region 0 by the HP CPU1 Peripheral bus.

0: Not monitor

1: Monitor

(R/W)

Continued on the next page...

Register 21.48. ASSIST_DEBUG_CORE_1_MONTR_ENA_REG (0x0000)

Continued from the previous page...

ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_ENA Configures whether to monitor write operations in HP CPU1 region 0 by the HP CPU1 Peripheral bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_ENA Configures whether to monitor read operations in HP CPU1 region 1 by the HP CPU1 Peripheral bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_ENA Configures whether to monitor write operations in HP CPU1 region 1 by the HP CPU1 Peripheral bus.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_ENA Configures whether to monitor HP CPU1 SP going below the lower bound address of HP CPU1 SP monitored region.
0: Not monitor
1: Monitor
(R/W)

ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_ENA Configures whether to monitor HP CPU1 SP going beyond the upper bound address of HP CPU1 SP monitored region.
0: Not monitor
1: Monitor
(R/W)

Register 21.49. ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MIN_REG (0x0090)

31 … 0	
ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MIN	
0xffffffff	Reset

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MIN Configures the lower bound address of HP CPU1 Data bus region 0. (R/W)

Register 21.50. ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MAX_REG (0x0094)

31 … 0	
ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MAX	
0	Reset

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_MAX Configures the upper bound address of HP CPU1 Data bus region 0. (R/W)

Register 21.51. ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MIN_REG (0x0098)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MIN	
31 ... 0	
0xffffffff	Reset

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MIN Configures the lower bound address of HP CPU1 Data bus region 1. (R/W)

Register 21.52. ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MAX_REG (0x009C)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MAX	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_MAX Configures the upper bound address of HP CPU1 Data bus region 1. (R/W)

Register 21.53. ASSIST_DEBUG_CORE_1_AREA_PIF_0_MIN_REG (0x00A0)

ASSIST_DEBUG_CORE_1_AREA_PIF_0_MIN	
31 ... 0	
0xffffffff	Reset

ASSIST_DEBUG_CORE_1_AREA_PIF_0_MIN Configures the lower bound address of HP CPU1 Peripheral bus region 0. (R/W)

Register 21.54. ASSIST_DEBUG_CORE_1_AREA_PIF_0_MAX_REG (0x00A4)

ASSIST_DEBUG_CORE_1_AREA_PIF_0_MAX	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_1_AREA_PIF_0_MAX Configures the upper bound address of HP CPU1 Peripheral bus region 0. (R/W)

Register 21.55. ASSIST_DEBUG_CORE_1_AREA_PIF_1_MIN_REG (0x00A8)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_MIN	
31 ... 0	
0xffffffff	Reset

ASSIST_DEBUG_CORE_1_AREA_PIF_1_MIN Configures the lower bound address of HP CPU1 Peripheral bus region 1. (R/W)

Register 21.56. ASSIST_DEBUG_CORE_1_AREA_PIF_1_MAX_REG (0x00AC)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_MAX	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_1_AREA_PIF_1_MAX Configures the upper bound address of HP CPU1 Peripheral bus region 1. (R/W)

Register 21.57. ASSIST_DEBUG_CORE_1_AREA_PC_REG (0x00B0)

ASSIST_DEBUG_CORE_1_AREA_PC	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_1_AREA_PC Represents the HP CPU1 PC value when an interrupt is triggered during region monitoring. (RO)

Register 21.58. ASSIST_DEBUG_CORE_1_AREA_SP_REG (0x00B4)

ASSIST_DEBUG_CORE_1_AREA_SP	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_1_AREA_SP Represents the HP CPU1 SP value when an interrupt is triggered during region monitoring. (RO)

Register 21.59. ASSIST_DEBUG_CORE_1_SP_MIN_REG (0x00B8)

ASSIST_DEBUG_CORE_1_SP_MIN	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_1_SP_MIN Configures the lower bound address of the HP CPU1 SP monitored region. (R/W)

Register 21.60. ASSIST_DEBUG_CORE_1_SP_MAX_REG (0x00BC)

ASSIST_DEBUG_CORE_1_SP_MAX	
31 ... 0	
0xffffffff	Reset

ASSIST_DEBUG_CORE_1_SP_MAX Configures the upper bound address of the HP CPU1 SP monitored region. (R/W)

Register 21.61. ASSIST_DEBUG_CORE_1_SP_PC_REG (0x00C0)

ASSIST_DEBUG_CORE_1_SP_PC	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_1_SP_PC Represents the HP CPU1 PC value during SP monitoring. (RO)

Register 21.62. ASSIST_DEBUG_CORE_1_INTR_RAW_REG (0x0084)

31 … 10	9	8	7	6	5	4	3	2	1	0	
(reserved)	ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_RAW	ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_RAW	ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_RAW	ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_RAW	ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_RAW	ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_RAW	ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_RAW	ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_RAW	ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_RAW	ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_RAW	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	0	Reset

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_INT. (RO)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_INT. (RO)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_INT. (RO)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_INT. (RO)

ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_INT. (RO)

ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_INT. (RO)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_INT. (RO)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_INT. (RO)

ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_INT. (RO)

ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_RAW The raw interrupt status of ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_INT. (RO)

Register 21.63. ASSIST_DEBUG_CORE_1_INTR_ENA_REG (0x0088)

(reserved)	ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_INTR_ENA	ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_INTR_ENA	ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_INTR_ENA	ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_INTR_ENA	ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_INTR_ENA	ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_INTR_ENA	ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_INTR_ENA	ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_INTR_ENA	ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_INTR_ENA	ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_INTR_ENA	
31 ... 10	9	8	7	6	5	4	3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	0	Reset

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_INT. (R/W)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_INT. (R/W)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_INT. (R/W)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_INT. (R/W)

ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_INT. (R/W)

ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_INT. (R/W)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_INT. (R/W)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_INT. (R/W)

ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_INT. (R/W)

ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_INTR_ENA Write 1 to enable ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_INT. (R/W)

Register 21.64. ASSIST_DEBUG_CORE_1_INTR_CLR_REG (0x008C)

Bits	31–10	9	8	7	6	5	4	3	2	1	0	
Field	(reserved)	ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_CLR	(ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_CLR	(ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_CLR	(ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_CLR	(ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_CLR	(ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_CLR	(ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_CLR	(ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_CLR	(ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_CLR	(ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_CLR	
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	0	Reset

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_RD_INT. (WT)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_AREA_DRAM0_0_WR_INT. (WT)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_RD_INT. (WT)

ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_AREA_DRAM0_1_WR_INT. (WT)

ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_AREA_PIF_0_RD_INT. (WT)

ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_AREA_PIF_0_WR_INT. (WT)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_AREA_PIF_1_RD_INT. (WT)

ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_AREA_PIF_1_WR_INT. (WT)

ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_SP_SPILL_MIN_INT. (WT)

ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_CLR Write 1 to clear ASSIST_DEBUG_CORE_1_SP_SPILL_MAX_INT. (WT)

Register 21.65. ASSIST_DEBUG_CORE_1_RCD_EN_REG (0x00C4)

	(reserved)	ASSIST_DEBUG_CORE_1_RCD_PDEBUGEN	ASSIST_DEBUG_CORE_1_RCD_RECORDEN	
Bits	31 … 2	1	0	
	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	Reset

ASSIST_DEBUG_CORE_1_RCD_RECORDEN Configures whether to enable HP CPU1 PC logging.

0: Disable

1: ASSIST_DEBUG_CORE_1_RCD_PDEBUGPC_REG starts to record HP CPU1 PC in real time

(R/W)

ASSIST_DEBUG_CORE_1_RCD_PDEBUGEN Configures whether to enable HP CPU1 debugging.

0: Disable

1: HP CPU1 outputs HP CPU1 PC

(R/W)

Register 21.66. ASSIST_DEBUG_CORE_1_RCD_PDEBUGPC_REG (0x00C8)

	ASSIST_DEBUG_CORE_1_RCD_PDEBUGPC	
Bits	31 … 0	
	0x000000	Reset

ASSIST_DEBUG_CORE_1_RCD_PDEBUGPC Represents the PC value at HP CPU1 reset. (RO)

Register 21.67. ASSIST_DEBUG_CORE_1_RCD_PDEBUGSP_REG (0x00CC)

ASSIST_DEBUG_CORE_1_RCD_PDEBUGSP	
31 ... 0	
0x000000	Reset

ASSIST_DEBUG_CORE_1_RCD_PDEBUGSP Represents the HP CPU1 SP. (RO)

Register 21.68. ASSIST_DEBUG_CORE_1_LASTPC_BEFORE_EXCEPTION_REG (0x00F0)

ASSIST_DEBUG_CORE_1_LASTPC_BEFORE_EXC	
31 ... 0	
0	Reset

ASSIST_DEBUG_CORE_1_LASTPC_BEFORE_EXC Represents the HP CPU1 PC of the last command before the HP CPU1 enters exception. (RO)

Register 21.69. ASSIST_DEBUG_CORE_1_DEBUG_MODE_REG (0x00F4)

(reserved)	ASSIST_DEBUG_CORE_1_DEBUG_MODULE_ACTIVE	ASSIST_DEBUG_CORE_1_DEBUG_MODE	
31 … 2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	Reset

ASSIST_DEBUG_CORE_1_DEBUG_MODE Represents whether HP CPU1 is in debugging mode.
1: In debugging mode
0: Not in debugging mode
(RO)

ASSIST_DEBUG_CORE_1_DEBUG_MODULE_ACTIVE Represents the status of the HP CPU1 debug module.
1: Active status
Other: Inactive status
(RO)

Register 21.70. ASSIST_DEBUG_CLOCK_GATE_REG (0x0108)

(reserved)	ASSIST_DEBUG_CLK_EN	
31 … 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	1	Reset

ASSIST_DEBUG_CLK_EN Configures whether to enable the register clock gating.
0: Disable
1: Enable
(R/W)

Register 21.71. ASSIST_DEBUG_DATE_REG (0x03FC)

(reserved)	ASSIST_DEBUG_DATE	
31 28	27 0	
0 0 0 0	0x2109130	Reset

ASSIST_DEBUG_DATE Version control register. (R/W)